Quantum tunnelling

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Version 1.0
This is the first official version.

Introduction

This document is part of a portfolio of documents for a research project titled “The Particle Zoo”. As what is presented here is sharply at odds with established theories and ideas, it is necessary to read the documents in a specific order, as the ideas presented build upon each other.
Before reading this document, it is highly recommended that the parent document titled: “Properties of particles v_1.pdf” is read and understood. Recommended is also this document: “Generation of a photon through spontaneous emission v_1.pdf”.

As this is an ongoing study, more documents and any updates of the existing ones will be added as they become available.

Summary

The calculations and discussions presented here have demonstrated or suggested the following:

- That quantum tunnelling can be explained without Heisenberg’s uncertainty principle.
- That tunnelling is due to refraction in the aether; this being the same principle that prevents electrons in atomic orbitals from radiating energy and thereby collapse into the nucleus.
- That there is no such thing as short-term “borrowing of energy” from a particle’s nearest environment, with a return of this energy shortly after.
- That there are no quantum principles involved and hence there is no quantum mystery.
- That tunnelling is yet another verification of the existence of an aether.

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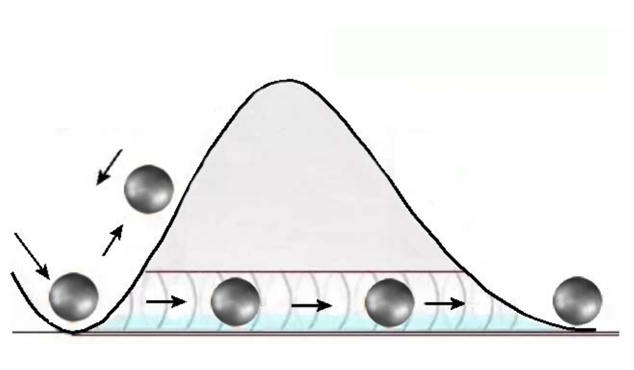
1 Quantum tunnelling

1.1 Introduction

Quantum tunnelling is one of the many phenomena under the umbrella of quantum mechanics. It is the phenomenon whereby a particle tunnels through an energy barrier, which it from a classical point of view should not be able to surmount. But as it seems, what happens at the quantum scale is in a realm of its own, with no parallel in the macro world. This introduces a certain element of strangeness and even mysticism – a projection of the inner workings of Nature, being something we just have to accept as facts.

Quantum tunnelling is believed to have an active role in the nuclear fusion in stars like the Sun. In modern electronics technology it has found widespread use, such as programmable devices, the tunnel diode, quantum computing and the scanning tunnelling microscope. It can be described very accurately in mathematical terms, but the level of understanding is based on phenomena that themselves are not fully understood.



The most common way of describing the principle of quantum tunnelling is with a rolling ball representing a particle with a certain amount of kinetic energy. This ball is met with a ramp that is sufficiently high to prevent the ball from rolling over the top and over to the opposite side. However, quantum tunnelling allows the particle through with some limited probability, as if it has succeeded in passing *through* the barrier itself and not rolling over it.

Classical mechanics predicts that a ball with insufficient energy to surmount a potential barrier will not be able to reach the other side. The potential barrier here is gravity. The ball will come to a halt and then start rolling back down, as illustrated above. If this scenario is considered in classical terms with particles like electrons instead of balls, they will bounce back if their kinetic energy is insufficient to protrude the barrier.

1.2 The quantum mechanical explanation

In quantum mechanics, atomic particles can, albeit with a very small probability, tunnel[1] through to the other side - seemingly by passing through the barrier, even if it has less energy than the barrier.

With some statistical probability, this act of particles suddenly appearing where they should not normally be, is explained in terms of Heisenberg uncertainty principle and the wave - particle duality of matter. In quantum mechanics, the uncertainty principle is any of several mathematical inequalities asserting a fundamental limit to the precision with which certain pairs of physical properties of a particle can be known. One such so-called complementary pair of variables is the position and momentum of a particle. Attempting to measure the exact position of a particle at any instant is tantamount to a correspondingly less precision in the measurement of its momentum, i.e. its speed. This defines an upper limit on how precisely these variables can be known at the same

[1] Note that the "tunnel" itself is a rather poorly defined analogy. There is no actual "tunnel" involved, just a probability of finding a particle somewhere.

time. In mathematical terms this is commonly expressed as an inequality $\sigma_x \sigma_p \geq \hbar/2$, where the product of the standard deviation of position σ_x and the standard deviation of momentum σ_p cannot be greater than $\hbar/2$, where $\hbar$ is the reduced Planck constant $h/2\pi$. This is a fundamental physical limit, which cannot be improved upon through refined measuring techniques.

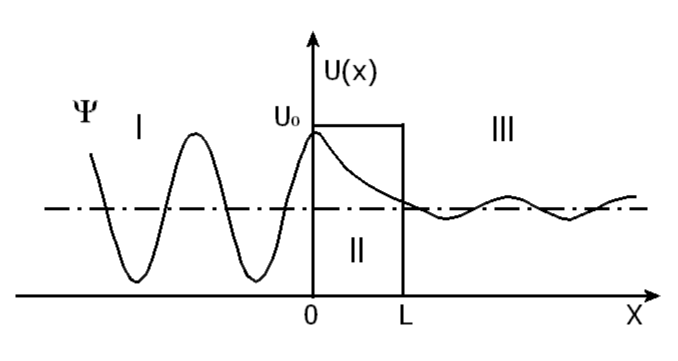
For example, if the exact calculation for its position was taken as a probability of 1, i.e. with absolute certainty, the standard deviation of the other variable – in this case the product of its speed and mass would have to be infinite; in practice having any possible speed. In between these extreme values for the standard deviations of this complementary pair there can be any combination of values, but only as long as the above inequality is satisfied. Hence, the probability of a given particle's presence on the opposite side of an intervening barrier is actually non-zero, and such particles will appear on the opposite side of the barrier at a rate proportional to this probability. All this of course is in stark contrast to a classical, everyday view.

In accord with the commonly accepted view of a wave - particle duality, all particles have wave functions. These wave functions are the solutions to the Schrödinger equation, which determine how particles act and interact.

As it is, a quantum has no exact location. Its location is a probability function until you measure it. This probability function is the square of the solution to the Schrödinger equation at each location. Quanta can surmount an insurmountable barrier as long as the probability is non-zero.

The above principle is illustrated graphically below, where an electron is coming in from the left, having an energy E < U_0. Here Ψ_n is the wave function of the electron, i.e. the solution to the Schrödinger equation for the regions I, II and III respectively. Since the wave function is non-zero in region III, some particles will succeed in penetrating the barrier, whereas most will be reflected.

As might be expected, as U_0 and L decreases, the probability of tunnelling goes up.



The wave function solutions are:

$$\Psi_I = A_1 e^{jk_1x} + A_2 e^{-jk_1x}$$
$$\Psi_{II} = B_1 e^{-jk_2x} + B_2 e^{jk_2x}$$
$$\Psi_{III} = C_1 e^{jk_1x} + C_2 e^{-jk_1x}$$
$$k_1 = \frac{\sqrt{2mE}}{\hbar} \quad \text{and} \quad k_2 = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

$C_2 = 0$ as there is no barrier on the right of region III to reflect off from.

Although the particle can have a non-zero probability to be in the potential barrier, that probability drops off very quickly over a distance that decreases as the (particle) mass increases and the difference in energy between the barrier and the particle increases. Consequently, the larger the mass and energy difference, the faster the probability decreases. Now, if the mass of the particle is small and the energy difference is small, then the particle can have a non-negligible probability of tunnelling. If the mass and energy difference are too large, as for the ball rolling up a hill, then the probability of tunnelling is negligible. This is said to be the reason why this quantum phenomenon is never noticed in everyday life. The phenomenon is present, but because of the relatively large mass of all bodies involved the resulting probabilities vanish.

The above is the mathematical description of it. How does this translate into an "under the hood" low-level physical description? The act of tunnelling itself is explained by the particle's apparent ability to borrow energy from its surroundings, and by means of this short-lived energy boost succeed in penetrating the barrier. If successful, it will shoot through the barrier. The amount of energy borrowed to achieve this is said to be "paid back" by making the reflected particles more energetic than they otherwise would have been. By this is presumably meant electrons that failed to borrow energy and hence got reflected with some extra kinetic energy. More about this below.

This is actually a rather extraordinary claim. How can a particle literally borrow energy from its surroundings and then somehow pay it back? In fact, the phenomenon of quantum tunnelling opens a range of questions that have no obvious answers. In denial of an aether medium, the aether is certainly not an energy source to borrow from. If the electron has the inherent ability to borrow energy from its surroundings, then why doesn't this happen all the time? With all electrons being absolutely identical, why should only a select few particles be given the privilege of an energy boost, whereas the great majority will not benefit from it and instead get reflected?

This is just one of the many quantum phenomena that is just as surreal and magical as it appears. Thus, we see that quantum mechanics and classical mechanics differ widely in their description of this scenario.

1.3 A new interpretation of quantum tunnelling

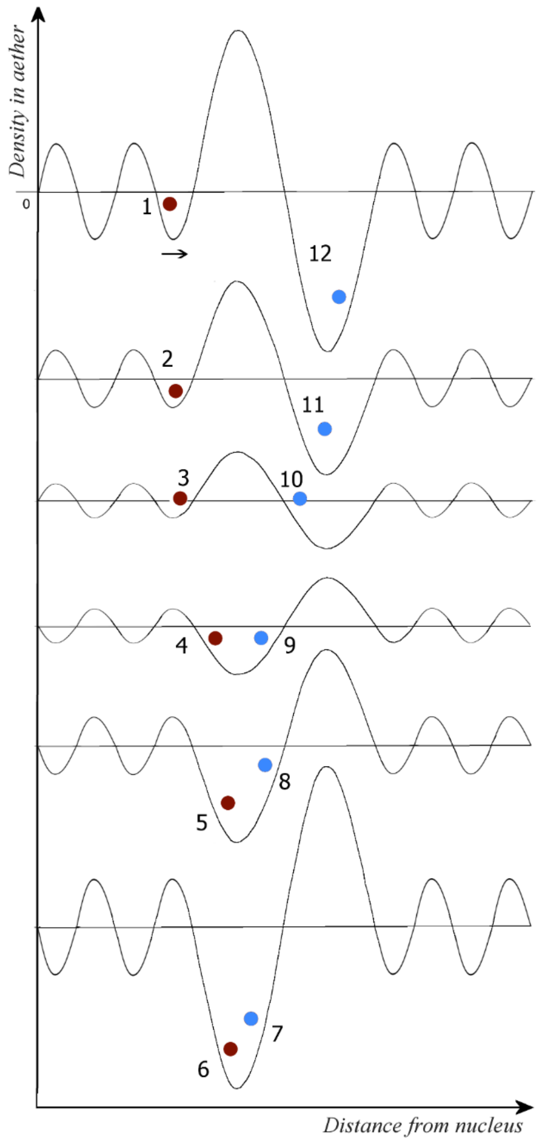
There is overwhelming evidence that the wave function gives a precise *numerical* account of the quantum tunnelling phenomenon. What we will present here is a rather different view of what takes place at the lowest physical level during such events. Perhaps both surprising and at the same time also not so very surprising, is that there is a great deal of commonality between what takes place in atomic orbitals and quantum tunnelling. We will illustrate this by the set of figures below, which at first glance looks quite similar to the one used for describing electron motions in their nuclear orbitals and spontaneous emission [2].

To recap from the referenced document: These wave patterns depict the aether energy *densities* at various successive instants in time when plotted against radial distance from the nucleus. The basic wave structure is that of longitudinal standing waves. The radial motion is caused by refraction due to the varying aether densities represented by the crests and troughs of the waves. Like a photon in a single mode optical fibre, the particle will veer away from where the refractive index is higher. This is at the crest of the longitudinal wave, as here the density of the aether wave is highest. Thus, the particle will follow naturally a path of minimum energy.

This is largely the situation also for the figure below, but here one will notice that at a particular distance from the nucleus a certain large impulse is overlaid; a spike which might have been caused by some sudden event in the atomic nucleus. This event might have been an instant of radioactive decay, with the emission of an alpha particle and an abrupt change in the otherwise regular density pattern as a result. Electron capture, whereby an electron is tunnelling into the nucleus is also possible This is known as inverted β^- decay, whereby an orbital electron is captured by the nucleus and combining with a proton to form a neutron and releasing a neutrino to balance the energy.

The coloured dots are approximations of the particle's position in the respective aether density wells as these vary over time and distance. The numbers 1 – 12 refer to specific situations on the plot.

[2] See document titled: "Generation of a photon through spontaneous emission v_1.pdf".



1. An electron, a proton or an alpha-particle ejected from the nucleus is trapped in a density well. As for spontaneous emissions, the particle will “bounce” between the refractive walls. This oscillation will take place at an immense rate. For hydrogen the time period for one complete cycle was found to be of the order of 10^{-22} s. Here the particle is approaching the region of an impulse caused by some event in the nucleus. This spike is not necessarily synchronized with the basic waves, neither in phase nor amplitude.
2. The waveforms are diminishing in height. The large spike is depicted as synchronous with the other wave crests, but again this may or may not be case.
3. The trough belonging to the spike is also diminishing. The particle has still got momentum out and away from the nucleus, which assists to a possible cross-over.
4. The particle had sufficient momentum to get across and is now embedded in the trough actually belonging to the large spike.
5. The particle is falling still deeper into the trough of the large spike.
6. The particle has reached the lowest possible position of density level.
   The oscillation will start reversing next.
7. Because the spike is here assumed to have a period twice that of the basic wave, the particle has still some way to move until it hits the refractive wall of the spike.
8. By having a longer wavelength as depicted here, the leading edge of the crest set up by the spike has therefore a smaller slope than the other wave crests.
9. The smaller slope does that the particle will find it easier to continue its motion in its outwards direction, this rather than bouncing back.
10. The particle has now succeeded in crossing half the width of the barrier.
    This would not have been possible had the phase relationship of the spike been such that its amplitude had remained high for a longer period.
11. The particle has now succeeded in passing through half the thickness of the barrier.
12. Depending on the shape of the large spike and the phase relations with the standing waves, the particle *may* succeed in passing through the remaining part of the barrier.

Note that these waveforms are for descriptive purposes only and do not purport to reflect the true waveforms. In particular, the spike might appear in countless shapes, both in terms of amplitude, width and content of harmonics. Here it is given the simplest possible shape of a sinusoidal impulse of limited duration, and a wavelength twice that of the waves caused by the nucleus.

Quite possibly only very few combinations of amplitude, phase and pulse width will constitute the right conditions for tunnelling, making this phenomenon the rare event it actually is. Contrary to what quantum theory asserts, all what happens is fully deterministic. It is the multitude of possible wave shapes of this spike and also its phase relationship with the basic waveform that makes quantum tunnelling *appear* as a statistically varying phenomenon.

Einstein was right –
"God does not play dice[3] with the Universe".

By the theory outlined above, the process involving borrowing of energy from the particle's surroundings might be due to the spike itself being the source of this energy. Returning to case 1 of the large illustration above, the phase and amplitude relations might be such that the particle will bounce a great many times within its present well before the conditions are present for passing through the barrier. Crossing the barrier is generally not a question of brute force. It is all about the right conditions – conditions that might first arise after many cycles due to the spike being asynchronous with the other waves.

With this theory one can easily see that there is no need to resort to Heisenberg's uncertainty principle as the physical reason for quantum tunnelling. The potential barrier is not static, as everything is waves. If considering the barrier as a static potential of a fixed height, the theory presented here would fail. Mathematically the uncertainty principle might be sound, but all it does is to cloak and mystify the real reason for the phenomenon.

Earlier in this chapter it was mentioned that particles that were not sufficiently energetic to cross the barrier would be reflected, and that some of these might even get some boost in their kinetic energy. This can also be understood in terms of this theory, without any reference to the dubious idea of energy being returned after a short-term "loan". The energy source is the spike. Again, depending on the relative phase and amplitudes between the spike and the basic wave at the location where the particle is located, the fast-rising edge of the spike might give the particle some extra kick if impacting this edge at an appropriate instant.

1.4 Some facts about quantum tunnelling

Although we in this theory have rejected tunnelling as a quantum effect, we shall continue to use "quantum tunnelling" interchangeably with just "tunnelling".

Quantum tunnelling is not restricted to nuclear physics, but was a general result of quantum mechanics that applies to many different systems. In 2016, the quantum tunnelling of water was discovered.

Quantum tunnelling occurs with barriers of thickness around 1-3 nm and smaller and is the cause of some important macroscopic physical phenomena. For instance, tunnelling is a source of current leakage in very-large-scale integration (VLSI) electronics and results in the substantial power drain

[3] Einstein liked a deterministic universe and loathed the randomness of certain aspects of quantum mechanics and didn't think it was real. He was not religious, and the "God" he is referring to is probably the Nature itself.

and heating effects that trouble high-speed and battery-powered technology. It is considered the lower limit on how small computer chips can be made.

Tunnelling is essential for nuclear fusion in stars. Temperature and pressure in the core of stars are believed to be insufficient for nuclei to overcome the Coulomb repulsion in order to achieve thermo-nuclear fusion. However, there is some probability to penetrate the barrier due to tunnelling. Although the probability is very low, the extreme number of nuclei in a star generates a steady fusion reaction over millions or even billions of years.

Radioactive decay is the emission of particles and energy from the unstable nucleus of an atom, resulting in a more stable product. This is done via the tunnelling of a particle out of the nucleus. This is considered the first application of the quantum tunnelling principle.

Proton tunnelling is a type of quantum tunnelling involving the instantaneous disappearance of a proton in one location and the appearance of the same proton at an adjacent location separated by a potential barrier. The two available positions are bounded by a double well potential, whereby its shape, width and height are determined by a set of boundary conditions.

The probability for a particle to tunnel is inversely proportional to its mass and the width of the potential barrier. Electron tunnelling is well-known. But as a proton is almost 2000 times more massive than an electron, it has a much lower probability of making a successful tunnelling. Nevertheless, proton tunnelling still occurs, especially at low temperatures and high pressures where the width of the potential barrier is small.

To conclude: There is nothing mysterious about quantum tunnelling. It got its aura of mysticism because its real cause has been denied – the existence of waves in the aether.

2 References

1. https://en.wikipedia.org/wiki/Quantum_tunnelling
2. https://en.wikipedia.org/wiki/Electron_capture
3. https://en.wikipedia.org/wiki/Alpha_decay
4. https://en.wikipedia.org/wiki/Proton_tunneling